

Using Participatory Design in the Development of the Homegrown Dashboard System for Farmer Collaboration and Contract Management

Lauren Forbes
University of Cincinnati
Cincinnati, Ohio, USA
lauren.forbes@uc.edu

Sai Meenakshi Hariharan
University of Cincinnati
Cincinnati, Ohio, USA
harihasi@mail.uc.edu

Annu Sible Prabhakar
School of Information Technology
University of Cincinnati
Cincinnati, Ohio, USA
annu.prabhakar@uc.edu

Abstract

Black-led urban farms provide a model of self-determination and resilience within low-income communities facing food insecurity in U.S. cities. In this paper, we describe the design of a dashboard-based collaboration system to support three Black urban farmers in a local food network (LFN) in a Midwestern city in the US. These farmers are contracted through the Homegrown study to provide fresh produce for a group-based, nutrition program for pregnant Black women. We leverage principles and practices of participatory design to iteratively co-create the dashboards with the farmers. This dashboard system provides a “proof of concept” for broader uses of the system beyond procurement, including farm and inventory management and collaboration across food systems actors within other similar LFNs. Given the lack of HCI research in Black-led urban farming, our study serves as a novel attempt to identify both unique challenges and opportunities for HCI researchers interested in applied research.

CCS Concepts

• Human-centered computing → Participatory design.

Keywords

Participatory Design, Urban Farming, Black-led Urban Farming, Local Food Network

ACM Reference Format:

Lauren Forbes, Sai Meenakshi Hariharan, and Annu Sible Prabhakar. 2025. Using Participatory Design in the Development of the Homegrown Dashboard System for Farmer Collaboration and Contract Management. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719682>

1 Introduction

Urban farming in the U.S. has become a commonplace activity practiced by people of all cultural and socio-economic backgrounds, despite its humble beginnings as a survival practice of the urban poor [1, 8]. Most urban farms are part of local food systems that

connect regional food producers to a market of food entrepreneurs, retailers, institutional purchasers, and consumers. These local food systems are embedded within complex, sometimes contentious socio-political contexts that shape the goals and priorities of its members. Differential access to resources (e.g., finances, land, training) between urban farmers within these systems can ultimately shape their ability to effectively collaborate and to achieve their objectives [6]. The study presented in this paper addresses some of the primary challenges that Black urban farmers face in their efforts to achieve their socially-driven objectives to create sustainable, autarkic food systems organizations and networks. This study was part of a larger study called *Homegrown: Cultivating Black Maternal Agency and Community Capacity for Healthy Futures* led by the first author. The Homegrown study created a closed-loop system of food supply and demand by linking the nutritional needs of pregnant Black women- a group that faces a disproportionately higher risk of experiencing poor birth outcomes than their counterparts- with the market access challenges experienced by Black urban farmers in a small Midwest city. The purpose of the pilot study presented in this paper was to co-design and develop a data management platform for participating farmers. This “dashboard system” was conceptualized, co-produced, and preliminarily tested to manage the administrative complexity of the Homegrown study; to facilitate data management capacity building among the farmer-vendors; and to support sustainable collaboration among the emerging farmer network. The Homegrown dashboard system includes a set of interactive, data dashboards along with corresponding data input forms and the servers and platforms on which they are stored and hosted. This system enables the collection and visualization of key process and outcome data, providing much-needed structure, record-keeping, and administrative processes for the product purchase contract of the Homegrown study. In this paper, we discuss how participatory design was used to collaboratively shape the dashboard system with end users and to facilitate clear communication across the multi-disciplinary research team involved. We discuss the opportunities of the Homegrown dashboard system and population-specific HCI design considerations that were identified through its development. We also provide insights into design considerations for Black farmers involved in HCI research, for maintaining institution-community partnerships, and for the effective management of interdisciplinary academic collaborations.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719682>

2 Background and Related Work

2.1 Black Farming and Food Sovereignty

Today, most historically segregated U.S. neighborhoods continue to struggle with the lasting effects of racialized poverty and ongoing economic divestment. Black urban farmers experience these effects firsthand as most are from, live in, and/or serve the residents of these marginalized communities. In resistance to the normalization of racialized inequity, these farmers actively work to counteract the institutional factors and cognitive frameworks that sustain economic and social inequities. They use food cultivation as a tool to rebuild the health and wellbeing of Black and low-income communities [4, 12]. However, these farmers face substantial barriers to the success and sustainability of their farms and the networks they participate in such as racially discriminatory bank lending practices, precarity of land tenure, and related administrative challenges [3, 8]. These barriers make it difficult for many Black farmers to acquire needed resources, to leverage emerging growth opportunities, and to secure, manage, and execute public food procurement contracts. The ability of these farmers to successfully achieve their food systems objectives depends largely on their access to these key resources, which enable them to build the administrative capacity necessary for sustainability and growth. Information technology can offer important tools and insights for these organizations, helping them to secure and manage scarce resources, facilitate partnerships, and utilize data to inform their decision-making processes. Understanding which IT solutions could work best and how to adapt them to meet the unique needs and opportunities of Black urban farmers requires careful consideration of the principles of HCI and participatory design.

2.2 Participatory Design, Institution-Community Partnership, and Data Visualization

While IT solutions offer many opportunities to meet the administrative and capacity building needs of Black urban farmers and the externally-funded projects (e.g., grants and contracts) in which many participate, consideration of the socio-cultural, economic, and political factors shaping both Black farmers' potential engagement patterns with IT tools and the paradigm of the developers building them is imperative both for product effectiveness and sustainable collaboration.

For example, we consider how the racialized historical context of research in the U.S. structures the institution-community relationships within this study. Many urban universities have a long and violent history of exploiting low-income Black communities in both research practices and in their urban expansion [10]. Resultingly, these myriad harms have led to deep mistrust of academic institutions and research among many Black populations [7]. This mistrust and ongoing institutional exploitation often makes it difficult to establish genuine rapport and mutual trust between researchers and Black communities. These relational dimensions are necessary to understand because they inform the perspectives and contexts in which participatory design occurs [3, 11].

Recognizing that our research team is, at all times, entangled within an institutional legacy of anti-Black violence and because of

the capacity limitations of our farmer-participants for continuous research activity participation, we necessarily tailored our approach to participatory design in this study. We do so by shifting some of the assumptions, priorities, and design considerations that inform our study. We leverage ongoing, community-based work of the first author and primary data collected in this study to develop a robust understanding of who Black urban farmers are holistically rather than relying on default, dominant culture archetypes of who an "urban farmer" is [5] and what their goals and needs are.

Moreover, data dashboards offer one option to address some of the administrative and capacity challenges of Black urban farmers while also presenting unique opportunities to facilitate trust building and power-sharing within institution-community partnerships. Dashboards are dynamic, web-based interfaces that are meant to "collect, summarize, and present information from multiple sources" using high-quality data visualizations [14]. Importantly, many dashboard platforms offer automatic data transferability and analysis features, a wide range of customizable visualizations, and platform integration capabilities, which vastly expand the functional possibilities of dashboards, enabling them to be much more than just a tool to systematize and support data-based decision-making and performance management. [14] distinguish functional and visual design features, noting that dashboard designs also communicate organizational values. Thus, dashboards can have both pragmatic and symbolic utility for organizations of all sizes.

Most of the scholarship on the public sector uses of dashboards comes from the healthcare, public health, and education sectors, where dashboards have been used to track population health [13], disease outbreak patterns [9], and student learning outcomes [2], respectively. Few studies exist that explore how dashboards are used within and across urban farms and even fewer studies link dashboard use to public contracts.

3 Method

We present the first part of the Homegrown dashboard system development, which occurred from summer 2023 to fall 2024. This is described here in three phases- *Needs Assessment*, which consists of semi-structured interviews, field work, and preliminary relationship building with local farmers who would later become vendors and participants in the Homegrown study; *Preliminary*, which discusses the development of low-fidelity dashboard prototypes and database design; and *Build Out & Beta-testing* which is the generation and testing of high-fidelity dashboard prototypes in the first cohort of the Homegrown study in late summer 2024. In addition to the interview data, we utilize dashboard prototype sketches, field notes gathered through informal interactions with the farmer community, and Homegrown study-related artifacts such as the supply chain distribution process and produce procurement list. This study received university IRB approval in summer 2023. Importantly, this study documents two co-occurring, related processes as phenomena of interest- that is, the design and development of the Homegrown dashboard system and the planning and implementation of the broader Homegrown study, which the dashboard system was originally designed to support. Our research team focused on the former study and includes three university affiliated researchers; the lead author, who is principal investigator of the Homegrown study and

leads the dashboard system development; the second author, an IT graduate student researcher at the time (now graduated) who built the system and led design feedback sessions; and the third author, an experienced HCI researcher. Each of our disciplinary backgrounds and cultural identities shapes the lens through which we approached this study.

3.1 Phase 1: Needs Assessment

We conducted semi-structured interviews with four local Black urban farmers in summer 2023 to gauge their interest in participating in the study and to understand how an "IT platform" (not yet conceptualized as a dashboard) could benefit them in both their farm activities and their ability to execute a complex procurement contract. These interviews were approximately one hour long and took place on-site at their respective farms. The number of farmers interviewed was determined by both the relatively small size of the local Black urban farming population and other socio-cultural considerations. For example, a set number of farmer-vendors was pre-determined for the contract, and interviews were intended to be a means of gauging both design considerations for the platform and interest of the interviewees in participating as a vendor. Ethically, the researchers determined that it was best to limit the interview pool to the maximum size of the number of farmers who would be directly contracted in the study; this would enable the information gathered to be maximally useful, and it would minimize the likelihood of a misunderstanding about a forthcoming contract offer. This decision was also made in consideration of the structure and nature of the relationships among the farmers in the network. Interviews were transcribed and analyzed thematically by the research team. The analysis focused on the identification of design considerations, challenges, and potential opportunities identified by the farmers that could be addressed by the IT platform. It also covered farmers' goals and thoughts about procurement contracts. Themes that emerged from the interviews highlighted a strong interest among the farmers in increasing the amount and quality of collaboration among local Black farmers and the barriers to doing so; the gap between existing levels of administrative capacity for and interest in information technology and the farmers' current use of information technology in managing their farms; and the centrality of Black cultural identity and community well-being as guiding forces informing both the purpose and strategies of their work. The farmers also expressed a strong desire for improving the effectiveness and sustainability of partnerships among farmers within their networks. They saw working together in a coordinated way as a means to both decrease redundancy and waste in their farm production. Collaboration, they believed, would provide a shared reward for all area farmers and the absence of such would continue to foster an unhealthy climate of competition for scarce resources. The following quotes illustrate findings from these interviews about collaboration goals, partnership intentions, and farm management processes. Quotes are anonymized due to the small number of farmers who participated in the interviews.

"It's like, why we all gotta be logging our stuff separately when we could be working together? If I know somebody else got them cucumbers on lock, I ain't gotta worry. I can focus on what I do best. We need a system where we can all see who's growing what

and where, so we ain't doubling up on the same crops. That's just wasting space and time."

"A farmer is taking his stuff, putting it in his truck, and going to a farmer's market. Most of these farmer's markets don't even exist in the neighborhoods that we're talking about need the help. So even if we want to create a farmer's market in [neighborhood1], or [neighborhood 2], there needs to be an easier way to communicate with the other urban farmers. To be able to say, "Hey, you know what? I've got 20 tomatoes over here that we're not gonna be able to get rid of. Does anybody have use for them?"

"I think that there's a huge benefit to that if all the growers can partner up and create enough of a working model for the work to be there, right?"

"We are struggling to figure out what land's free to plant and what ain't. It's like to find a needle in a haystack. We need something organized, something to tell us what's available so we can get our crops in the ground."

"Yeah, it's like to run a farm with one hand tied behind your back. We need some tech that's right there with us in the fields, you know? Like, if I could just pull out my phone and tell what needs picking, that'd save us a heap of time. We gotta keep up with the times, make farm management as easy as checking emails."

3.2 Phase 2: Preliminary System Development

3.2.1 Low Fidelity Prototype and Database Development. We began with "paper prototyping" through a series of hand-drawn sketches to illustrate the front-end layout of the dashboards and their design specifications. These were made using Samsung Notes on a tablet device and were iterated over time with a focus on data clarity, usability, and collaboration among end users. The sketches provided the structure and layout of a user-friendly, lightweight system that would facilitate the planning, management, and execution of farmers' produce procurement responsibilities according to the Homegrown contract. A set of low-fidelity dashboard prototypes were made to correspond with roles observed within the local food system, specifically food producers (local and regional), food distributors, food *re*-distributors, (those who aggregate and distribute unsold produce from food retailers), value added producers and retailers. A "Commons" dashboard was also designed as a shared space where all local food systems actors could meet virtually for information and data sharing.

Importantly, the dashboards and their composite visualizations were developed with the understanding that the Homegrown procurement contract is situated among a wider array of future contracts, collaboration opportunities, and growing seasons that shape the planning and decision-making context of the farmers. For this reason, the dashboard system was designed with the "bigger picture" goals, needs, and desires of the farmers in mind (as shared in the interviews), which extended beyond the immediate needs of the study. For example, a site mapping and land access feature was designed within the dashboards that could overlay farmers' existing growing sites against a map of all available vacant land parcels in the city. The Common dashboard was also designed to facilitate better collaboration and coordination across the growers as was a priority expressed in the interviews. To address this, preliminary plans for the dashboards included a farm information

look-up system, a standardized produce price list and unit quantity measurement system, a shared farm tool exchange and farm volunteer registry system, and a shared event calendar so farmers could support and promote each other's community events. These attributes reflected the expressed collaboration goals of the farmers that were interviewed. Moreover, in these digital sketches we paid particular attention to the design layout, color consistencies, variety versus utility balance in our selection of data visualization types (we prioritized streamlined, broadly familiar chart types), and clustering of data spatially within the dashboard layout based on data subcategories. We included filtering options to isolate data by time, farm, cohort, and crop type. We also used the tooltip feature to display additional information so that only necessary information was immediately visible in order to minimize end user data overload. All of these design choices were intended to make the dashboard layout maximally engaging, practical, and easy-to-use for farmers. These specifications were a direct response to the farmers' expressed prioritization of system usability along with the importance of avoiding the creation of additional labor for them to do through the system.

3.2.2 Matching System to Homegrown Study Development. As the low-fidelity prototypes and Excel spreadsheet database were being developed, the details of the Homegrown study were continually evolving, which presented unique challenges. Two key documents linked the two core parts of the study, the Local Food Network (LFN) and the Nutrition and Education Program (NEP)- the produce procurement schedule and the flow chart diagram illustrating the supply chain logistics. The former document detailed the specific produce items, item quantities, and program weeks in which those items were needed from each participating farmer. The latter was a diagram indicating the individuals, times, dates, data forms, and locations involved in produce transfer from food producer to end consumer. Both of these documents went through numerous iterations as both vendor availability and Homegrown NEP participant enrollment fluctuated, which affected the prototyping process. Further, farmer feedback from phase 1 indicated support for a process where each farmer in the LFN would be responsible for all of a single crop. We designed filters and easily viewable key information (e.g., current cohort number, delivery week, and produce items/quantities they were responsible for) to be central aspects of the vendor dashboards to make the system and study objectives more comprehensible. This included the Git-hub based HTML forms that we designed for data input on the produce quality, procurement handoffs, and farm management (e.g. planting and harvesting data). These forms provided a user-friendly interface for data input that farmers and our research team could use as needed. Through the supply chain and produce list documents, we identified three key points in the produce distribution process where data collection forms were needed, color-coding them as a memory heuristic and for diagram-to-data form correspondence. These forms included "Aggregation" for the initial aggregation of all produce from farmers (blue), "Receipt" for the hand off of aggregated produce to the research team (green), and "Distribution" for the assembly and distribution of produce bags to end consumers (pink). To minimize the workload on farmers, we

planned for the Aggregation form to be completed by the initial distributor (a vendor contracted to pick-up and transport the produce from one farmer to another), while the Receipt and Distribution Forms would be completed by a student research assistant. It soon became clear that incorporating contract payment information into the access-restricted farmer dashboards would be very useful as the contract payment details became increasingly complex to navigate. At this point, our initial vision of the dashboard system as an innovative research and administrative tool that could exist beyond the Homegrown study became clearer. Unlike other dashboards, the Homegrown dashboard system that our team was building had the potential capacity to facilitate greater transparency, accountability, and power-sharing within similar institution-community partnerships. With these broader equity-based objectives in mind, the team designed the Homegrown dashboard system with greater attention to future system replicability in other contexts. This parallel replicable system, essentially a skeleton of the functional one, became referred to as the "Homegrown dashboard shell".

3.3 Phase 3: Development & Beta Testing

High fidelity dashboard prototypes and revised data input forms were generated through ongoing dialogue with the farmers. Time and capacity constraints ultimately led us to focus on two key dashboards in this study- the farmer dashboard for individual farm-vendor, which supports the administrative and inventory management, and the shared "Common" dashboard. Quality control features were built into the dashboard system to ensure that produce quality was maintained; these consisted of a photo upload feature within the input forms and a rating assessment of produce quality. Two design feedback sessions were ultimately conducted- one in February 2024 where two farmers were involved, and one in spring of 2024 with one farmer. Of the original 4 farmers interviewed, 3 agreed to participate as vendors, and two 2 were ultimately contracted. This decrease in participation was due to participation challenges in recurring meetings set up to plan for vendor responsibilities in the study. As Spring 2024 approached, the bi-monthly meeting schedule became unsustainable for some participants. The onset of the growing season requires farmers to do many tasks to get ready for on-time planting; this is especially true for small urban farmers, many of whom are largely reliant on volunteers. Thus, participation in the research dimensions of this study (at least the meeting attendance component and participatory design of the system) presented challenges, and information discussed at the meetings was cumulative so any missed meeting not only limited one's capacity to contribute to decisions in future meetings, but it also challenged the group's ability to proceed effectively in collaborative planning. We also returned to the use of the Mural platform in this phase, which was initially used informally for diagramming the LFN in the region in Phase 1. Mural was used as an interdisciplinary communication and visioning tool, which supported the evolution of the Homegrown dashboard from a contract management tool for the Homegrown study to the prospective dashboard system shell for disseminated use among other emergent food networks. A dashboard "metasystem" framework was conceptualized to articulate the interconnected administrative, legal, and IT components and relationships within this system to inform

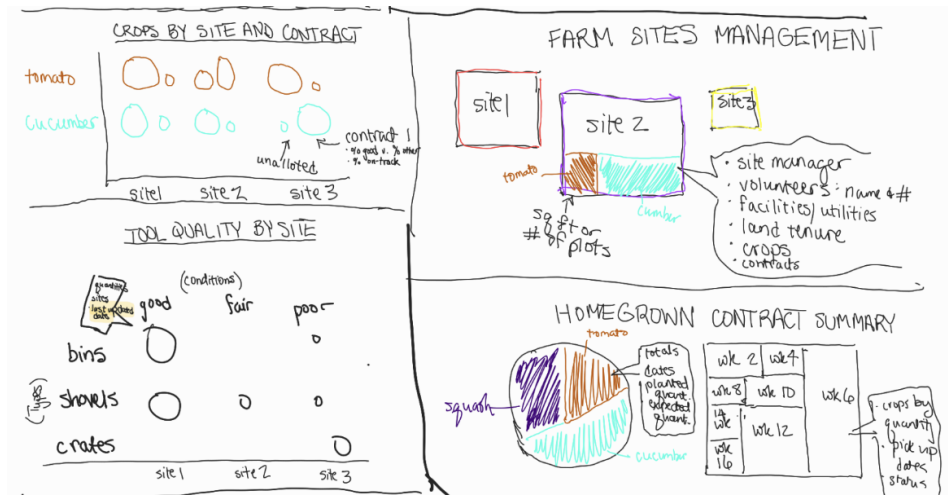


Figure 1: Visual examples from prototype sketches facilitating clear, interdisciplinary communication

	Week 1	Week 3	Week 5	Week 7	Total Per Cohort	Grand Total (Across 4 Cohorts)
Tomato (plum; roma)	25lb	25lb	15lb	50lb	115lb	460lb
Yellow Onion	50#	90#	90#	50#	280#	1120#
Red Onion	0	50#	0	50#	100# (50lb)	400#
Garlic	30#	50#	25#	25#	130#	520#

Figure 2: Abridged Homegrown study produce table used to generate vendor-specific produce procurement lists

the research-to-practice translation process. Moreover, Mural was also leveraged as a nascent method in this study. Recognizing the increasingly busy schedules of the farmers and their inability to meet regularly for participatory design of the dashboards, we developed a Mural dashboard diagram to reflect the dashboard layout and components envisioned based on the low-fidelity prototypes. Our team worked towards building this "high-fidelity diagram" into the Tableau-based high-fidelity prototype that was developed. This Mural was intended to be a shared "living" diagram that would be iterated and commented on by the farmers *on their own time* as the high-fidelity prototype development progressed. The diagram was embedded into the study website and was accessible on the same page as the Tableau dashboards with instructions on how to use it. In this way, the Mural diagram was intended to be a space for ongoing design iteration and co-production between the system designer and the farmers, thus allowing for continuous, open-ended feedback and inclusion of farmers' ideas and concerns, while also being considerate of their time and capacity constraints. This format of collaboration would also enable our team to avoid scheduling and logistical constraints and to track changes to the diagram (and corresponding dashboards) overtime. Ultimately, this planned Mural method was not utilized as the Homegrown study in its original form ended in fall 2024.

4 Discussion

The Homegrown dashboard system was developed to address the administrative complexities, data management needs, and sustainable collaboration objectives of Black urban farmers who were contracted as vendors in the Homegrown study. The iterative process of collaboratively developing this dashboard system from conceptualization to beta testing provides a unique context through which HCI principles and participatory design were applied. Key findings highlight the necessity of intentional alignment between administrative and technical design details and the importance of balancing design specifications of consistency and systematization with those of system adaptability and transferability. This tool, the Homegrown dashboard system, though still in early development, was created to address the administrative challenges of a complex pilot study and ensure effective procurement contract fulfillment. However, the early use of the dashboard system shell- the transferrable and adaptable components of the system for use in other institutional contexts- illustrates the utility of this tool to support broader collaboration, effective partnerships, and self-determination objectives of Black urban farmers.

Additionally, this study points to the immense value of in-person interaction between developers and study participants within the research process. The interactions Author 2 had with the farmers early in the dashboard development process, facilitated her ability to

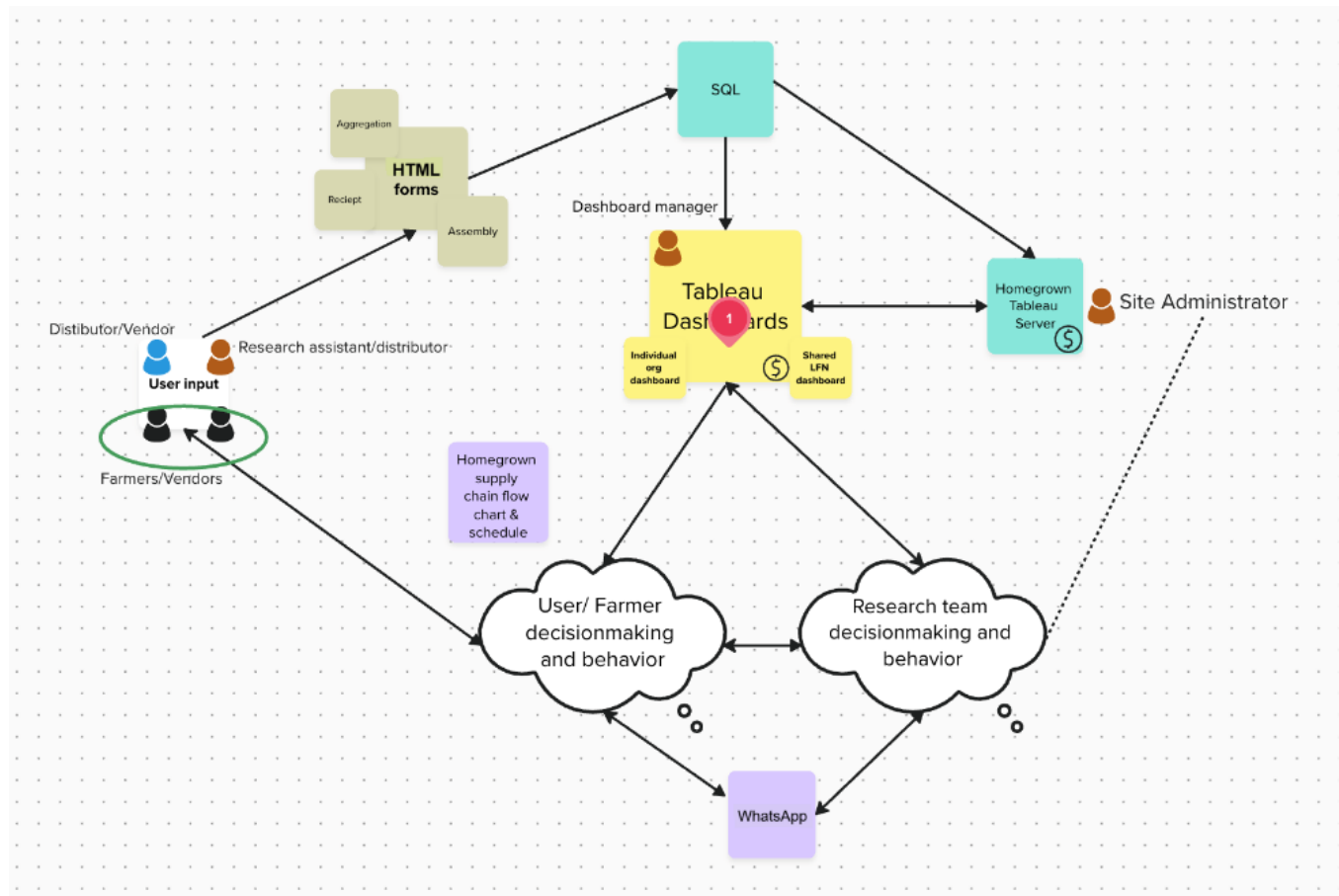


Figure 3: Mural diagram used by Author 1 to model the dashboard system technology and administrative components

understand the farmers' priorities and the context (socio-historical, political, cultural, and economic) that they operate within. Author 1's early engagement with these organizations facilitated rapport building which was essential for using participatory design in this study and for navigating study challenges that arose along the way. This type of direct interaction between developers and the individuals who use their products is not common for IT developers, and it facilitated the team's ability to challenge the dehumanizing process that often occurs when individuals involved in research are regarded only as "research subjects". These face-to-face interactions also supported the participants' ability to humanize Author 2 as an IT developer and to trust the research team enough to participate in the study. Thus, these interactions facilitated effective use of participatory design by fostering bridge-building within the partnerships involved and by personalizing both entities.

Other key lessons learned from this study were revealed in the limitations. The primary limitation we faced was the small number of farmer-vendors recruited for the study. This was largely due to the small size of the eligible population in our location, the one-year time commitment required for the procurement contract (reflecting the study length), and the limited funding available to cover vendor compensation, which is why there was a limited number

of vendors. The collective effect of these challenges was to limit the scope of information that could be gathered from both the design feedback sessions and beta testing; however, data gathered through the interviews were supplemented by Author 1's ongoing community-based work with these participants and other similar groups of farmers. Phase 4 (the current phase) of the Homegrown dashboard system development is currently underway through a new programmatic and partnership context. We are currently adapting the Homegrown dashboard system for use within a multi-year, farmer's market evaluation for a county health department and a grassroots, social impact community organization that manages the market. This institution-community partnership provides an important new collective action context in which we are utilizing participatory design to iteratively customize the Homegrown dashboard system and thereby determine its utility for supporting other local food system related contracts. The Homegrown dashboard system has likewise generated interest from other institutional and community actors representing local food systems with similarly, socially-driven objectives.

Through this innovative study, we contribute to the HCI and participatory design literature by identifying design specifications that are important to consider in the development of digital systems

for use within Black urban farming communities and institution-community partnerships. These include the time and capacity constraints that necessarily shape the research process; flexible, non-linear engagement of research participants with the participatory design process; the importance of having anticipatory structures, processes, and contingency mechanisms that can adapt to last minute system changes (e.g., produce quantity adjustments and emergency scenarios); the value of continuous reflexivity, cultural proximity, and relationship building of the research team with community partners; and the simultaneity of designing a product for both a specific case (e.g., Homegrown contract execution) and one adaptable for use in different scenarios with similar administrative contexts (e.g., the dashboard system shell). Additionally, the ongoing participatory design processes used to engage Black farmers in the development and testing of the dashboard system ensures that the perspectives, priorities, and preferences of the intended end users of this system are centered within the development process and not merely superficially engaged. Given that most HCI research and agriculture-related IT is informed by frameworks and methodologies derived from the dominant culture, we present a novel system that advances both diverse perspectives within HCI scholarship and equitable public administration. We look forward to engaging with the CHI community to discuss ways of advancing this work further.

This research is supported by funding from the University of Cincinnati Office of Research.

doi:10.1016/j.accinf.2011.08.002

References

- [1] Alison Hope Alkon. 2011. *Cultivating Food Justice: Race, Class, and Sustainability*. MIT Press.
- [2] Robert Bodily, Tarah K Ikahihifo, Benjamin Mackley, and Charles R Graham. 2018. The design, development, and implementation of student-facing learning analytics dashboards. *Journal of Computing in Higher Education* 30, 3 (2018), 572–598.
- [3] Lauren Forbes. 2023. Rooted resistance: The struggle for Black liberation through food cultivation. *Journal of Race, Ethnicity and the City* (2023), 1–27.
- [4] Hanna Garth and Ashanté M Reese. 2020. *Black food matters: Racial justice in the wake of food justice*. U of Minnesota Press.
- [5] Julie Guthman. 2008. Bringing good food to others: Investigating the subjects of alternative food practice. *Cultural Geographies* 15 (2008), 431–447. Issue 4.
- [6] Megan Horst, Nathan McClintock, and Lesli Hoey. 2017. The Intersection of Planning, Urban Agriculture, and Food Justice: A Review of the Literature. *Journal of the American Planning Association* 83 (2017), 277–295. Issue 3. doi:10.1080/01944363.2017.1322914
- [7] Kimberly D Manning. 2020. More than medical mistrust. *The Lancet* 396, 10261 (2020), 1481–1482.
- [8] Kristin Reynolds and Nevin Cohen. 2016. *Beyond the kale: Urban agriculture and social justice activism in New York City*. Vol. 28. University of Georgia Press.
- [9] Annett Schulze, Fabian Brand, Johanna Geppert, and Gaby Fleur Böhl. 2023. Digital dashboards visualizing public health data: a systematic review. doi:10.3389/fpubh.2023.999958
- [10] Robert Mark Silverman, Henry Louis Taylor Jr, Li Yin, Camden Miller, and Pascal Buggs. 2020. Are we still going through the empty ritual of participation? Inner-city residents' and other grassroots stakeholders' perceptions of public input and neighborhood revitalization. *Critical Sociology* 46, 3 (2020), 413–428.
- [11] Monica M White. 2011. D-Town Farm: African American Resistance to Food Insecurity and the Transformation of Detroit. *Environmental Practice* 13 (2011). Issue 4.
- [12] Monica M White. 2018. *Freedom farmers: Agricultural resistance and the Black freedom movement*. UNC Press Books.
- [13] Jue Wu, Kinnari Atit, Kay E Ramey, Grace Ann Flanagan-Hall, Mark Vondracek, Kemi Jona, and David H Uttal. 2021. Investigating students' learning through co-designing with technology. *Journal of science education and technology* 30 (2021), 529–538.
- [14] Ogan M. Yigitbasioglu and Oana Velcu. 2012. A review of dashboards in performance management: Implications for design and research. *International Journal of Accounting Information Systems* 13 (3 2012), 41–59. Issue 1.